Customer Behavior Analysis

End-to-end data analytics project integrating Python, SQL Server, and Power BI to extract actionable insights on customer purchasing patterns, segmentation, and revenue drivers.

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Project Objectives

Data Preparation

Clean and prepare raw transactional data for analysis

Exploratory Analysis

Perform EDA to understand patterns and trends

SQL Queries

Answer key business questions using structured queries

Interactive Dashboard

Build visualizations for stakeholder reporting

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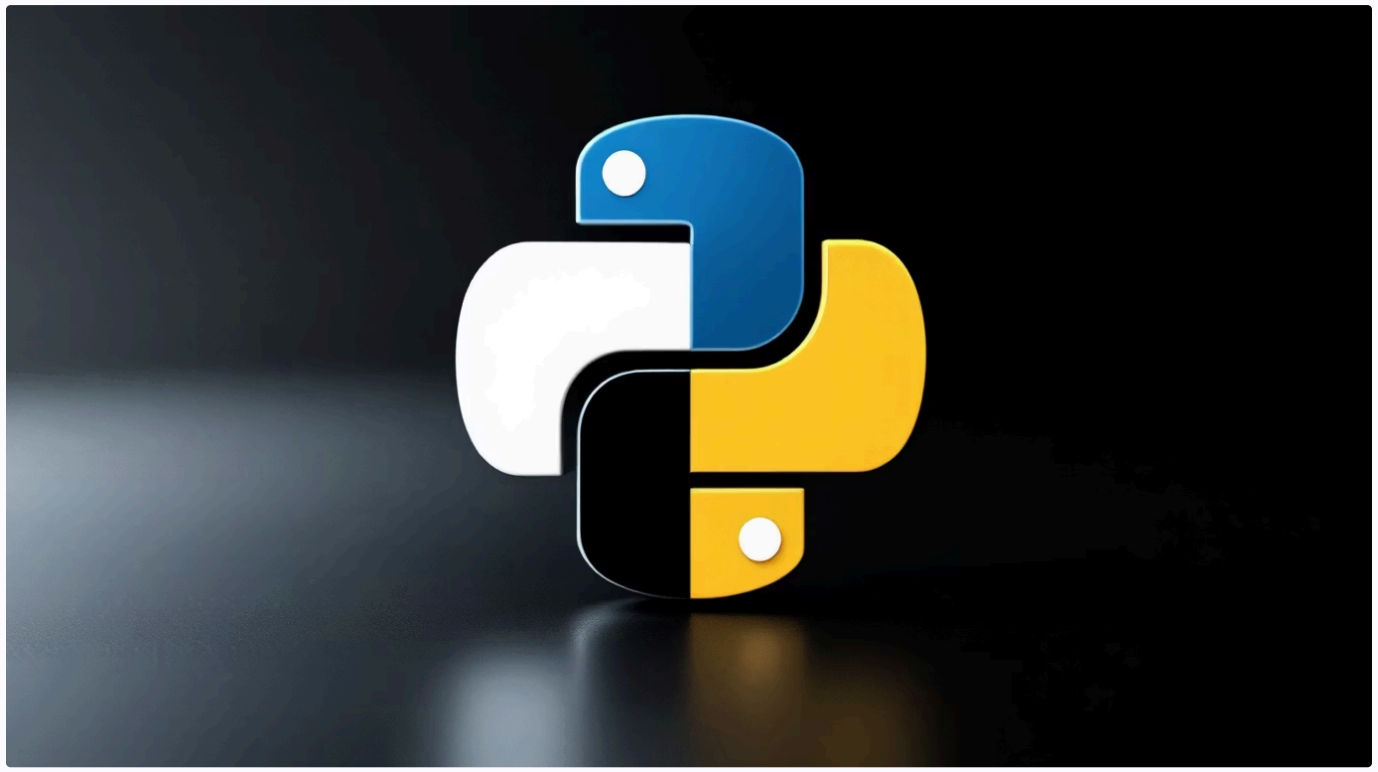
Tools & Technologies



Python Ecosystem

Pandas, NumPy, Matplotlib/Seaborn for data cleaning and EDA in Jupyter Notebook

SQL Server

Structured querying and business analysis layer

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Data Preparation Phase

Data Cleaning

Handled missing values, standardized categories, removed duplicates, ensured correct data types

Feature Engineering

Created purchase totals, segmentation indicators, discount flags for analysis

Exploratory Analysis

Examined distributions, trends, demographics, and subscription patterns

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SQL Analysis: Business Questions

Structured analysis of 10 key business questions using aggregations, window functions, and conditional logic.

01

Revenue by Gender

Male vs female customer comparison

02

Discount Users

Customers who used discounts above average

03

Top Products

Products by highest average rating

04

Shipping Comparison

Standard vs express shipping analysis

05

Subscription Behavior

Spending patterns by subscription status

06

Discount Usage

Products with highest discount percentage

07

Customer Segmentation

New, returning, and loyal customers

08

Category Leaders

Top 3 products per category

09

Repeat Buyers

Relationship between repeats and subscriptions

10

Age Groups

Revenue contribution by demographic

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Power BI Dashboard

3.9K	$59.76	3.75
Total Customers	Average Purchase	Average Rating

Interactive dashboard with filters for subscription status, gender, category, and shipping type.

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Key Insights

Customer Behavior

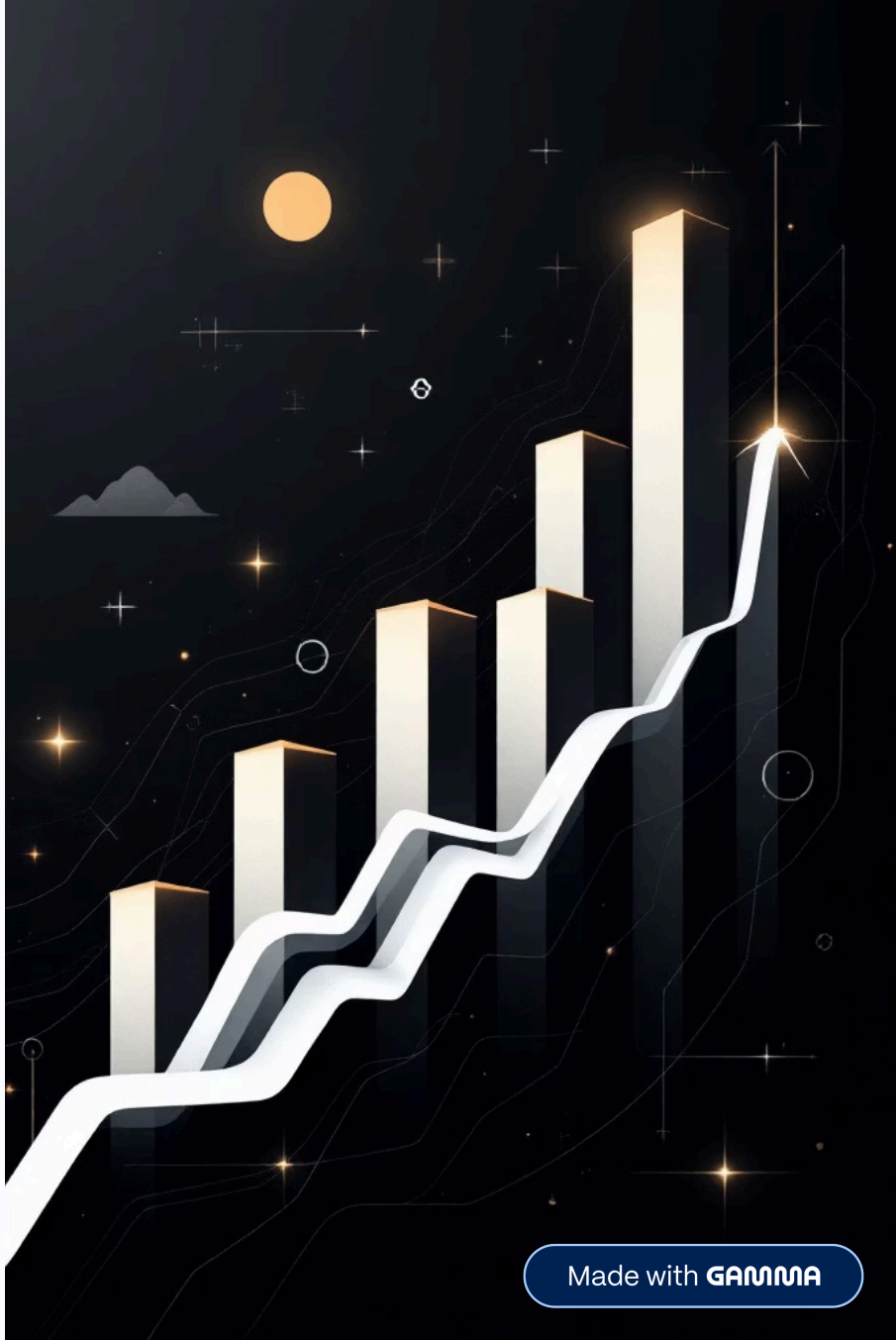
73% non-subscribed customers present growth opportunity

Revenue Drivers

Clothing dominates (~104K), young adults most valuable demographic

Purchasing Patterns

Discount usage doesn't reduce revenue; repeat customers show higher engagement



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Business Opportunities

Increase Subscriptions

Boost subscription incentives to convert non-subscribers

Focus Marketing

Concentrate on high-performing categories: Clothing and Accessories

Retention Strategies

Target young adults for improved customer retention

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Challenges & Solutions

Technical Challenges

- Data inconsistencies during cleaning
- Schema alignment between Python and SQL Server
- Accurate aggregation across dimensions

Design Challenges

- Dashboard clarity vs detail balance
- Dynamic filter implementation
- Visual hierarchy optimization

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Conclusion & Future Improvements

Complete data analytics pipeline demonstrating end-to-end project execution from raw data to actionable insights.

Current Success

Transformed data into insights, answered business questions, delivered interactive dashboard

Cloud Deployment

Deploy to Power BI Service for broader access

ML Integration

Implement machine learning for customer prediction

Real-Time Pipeline

Build streaming data infrastructure

Advanced Segmentation

Apply clustering techniques for deeper insights

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